AI Applied to Football



Pep Guardiola



Jürgen Klopp



Roberto De Zerbi

Knowledge Informed Playing Style Modelling and Explainable Manager Player Compatibility

Final Project Report

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1. Motivation and Problem Statement

1.1 The business problem

Every season, football clubs spend enormous sums signing players who never fit the way their team is meant to play. A technically gifted passer arrives at a club that presses aggressively and counters at speed, and never settles. A tireless, high energy runner is bought by a manager whose entire philosophy is built around patient positional possession, and ends up marginalised. These failures are not always about talent. They are about tactical fit, and tactical fit is notoriously hard to quantify. Recruitment departments still rely heavily on video analysis, scouting reports, and the intuition of experienced staff. All of this is valuable, but it is also slow, subjective, and difficult to scale across the hundreds of players a club might consider in a single transfer window.

At the same time, football has become one of the most heavily instrumented sports in the world. Every top division match in Europe now produces detailed statistical records covering passing, possession, defensive actions, and much more. Companies such as Opta and StatsBomb, and public mirrors of their data such as the one used throughout this project, have made this information available at a scale that would have been unimaginable a decade ago. The natural question that follows is whether this data can be used not just to describe what a player did, but to answer a more useful question for a club: given what we know about how a certain manager wants his team to play, how well would a specific player fit into that system, and why.

This project set out to answer exactly that question for three real, contemporary managers whose footballing identities are widely discussed among analysts and fans alike, Pep Guardiola, Jurgen Klopp, and Roberto De Zerbi. These three were chosen deliberately because they are commonly described as representing distinct footballing philosophies, even though, as this report will show, the truth is considerably more nuanced than the popular narrative suggests.

1.2 The research question

The central research question guiding this entire project is the following. Can football domain knowledge and player performance data be combined with machine learning to model the playing philosophies of elite football managers, and can that combined model estimate, in a way that can be explained in football language rather than in statistical jargon, how well an individual player fits each manager's footballing identity.

This question has two halves, and both matter equally. The first half is about modelling. It asks whether a manager's tactical identity, something that is normally described in qualitative terms by journalists and analysts, can be captured as a measurable, comparable profile built from real match data. The second half is about explanation. It asks whether the resulting compatibility assessment can be communicated in terms a football person would actually use and trust, rather than as an opaque number produced by a model nobody can interrogate.

1.3 Why this is hard, and why it is worth doing anyway

There are good reasons why nobody has fully solved this problem already. Managerial style is expressed through team structure, player positioning, timing, and coordination, things that are only partially visible in the kind of statistics that are publicly available. Two teams can produce very similar passing numbers while playing in completely different ways, because the numbers describe outcomes, not intentions. On top of that, a manager's statistical footprint is deeply entangled with the quality of the players available to him and the strength of the league he competes in. A team that dominates possession because it employs excellent passers is not necessarily doing so because of a particular philosophy. It might simply be better than most of its opponents.

Despite these difficulties, the project is worth doing for three reasons. First, even a partial, honestly evaluated answer to this question is more useful to a club than no answer at all, provided the limitations are stated clearly rather than hidden. Second, building the pipeline required to even attempt this question forces a rigorous confrontation with real world football data, including all of its gaps, inconsistencies, and traps, which is itself a valuable exercise for anyone working in football analytics. Third, and this turned out to be the most important lesson of the whole project, the process of trying and failing, then diagnosing why, then improving, produced a far more defensible and interesting result than either an easy success or a simple failure would have.

1.4 Objectives

The specific objectives set for this project were the following.

To build a verified, source checked record of exactly when each of the three managers was in charge of which club, at a level of precision that respects mid season changes rather than assuming a whole season belongs to one manager.

To audit, directly and empirically rather than by assumption, which public football data sources actually cover these three managers with sufficient detail, and to be transparent about what is and

is not available.

To build a football knowledge layer, meaning an explicit, documented set of football concepts and tactical dimensions that sit between raw statistics and any model, so that every number the system produces can be traced back to something a football person recognises.

To build profiles of each manager and of individual players on the same set of tactical dimensions, and to compute a compatibility score between any player and any of the three managers.

To make every compatibility score explainable, meaning that the system must be able to say which specific tactical characteristics drove a given score, not just report the score itself.

To evaluate the whole approach honestly, including actively trying to break it, rather than accepting a result that looked plausible without checking it.

To deliver a working, interactive application that lets a user browse real players from the top European leagues and see their compatibility with each of the three managers, with comparisons, explanations, and visualisations.

To document every methodological decision, every limitation, and every piece of evidence behind the project's conclusions, so that the work can be defended and, if needed, continued by someone else.

1.5 What this report covers

This report is organised so that it can be read from beginning to end as a complete account of the project, or consulted section by section as a reference. Section 2 provides the background a reader needs, both about the three managers and about the technical concepts used later. Section 3 reviews the existing academic literature and related systems, and positions this project's contribution relative to that literature. Section 4 is the longest section, and it walks through everything that was actually built, in the order it was built, including the data investigation, the knowledge layer, the modelling pipeline, and the application. Section 5 presents the results, including the honest account of what worked, what did not, and what was fixed. Section 6 closes with conclusions, stated limitations, and concrete recommendations for anyone who wants to take this work further.

2. Background

2.1 Playing style as a concept

Before any data can be collected or any model can be built, it helps to be precise about what is actually meant by playing style or tactical philosophy. In the football analytics literature, a team's playing style is generally understood as the repeated, characteristic pattern of behaviour a team shows both when it has the ball and when it does not. This is usually broken down into a small number of recurring themes that appear again and again across the academic literature and among professional analysts.

Build up refers to how a team begins its attacking possessions, particularly how involved the goalkeeper and defenders are in constructing play from the back rather than clearing the ball long. Progression refers to how a team advances the ball from its own half into more dangerous territory, whether through passing, dribbling, or a combination of both. Possession describes how much of the game a team spends with the ball, and how it retains it under pressure. Chance creation covers the mechanisms a team uses to generate shooting opportunities, such as crosses, through balls, or combination play in the final third. Pressing and counter pressing describe how aggressively a team tries to win the ball back, and how high up the pitch it does so. Transitions describe what happens immediately after possession changes hands in either direction, since the moments right after winning or losing the ball are often the most dangerous in the game. Defending covers the shape and behaviour a team adopts when it does not have the ball, including how high its defensive line sits and how compact its block is.

These themes are not independent of each other. A team that presses very high, for example, is also making a statement about how much space it accepts behind its defensive line, and a team that builds patiently through the goalkeeper is also making a statement about how comfortable it is inviting pressure. Understanding these interconnections is part of what it means to understand a manager's philosophy, and it is why this project treats football knowledge as something that should shape every stage of the analysis, not just something to mention in a discussion section at the end.

2.2 The three managers

2.2.1 Pep Guardiola

Pep Guardiola is widely regarded as the most influential exponent of what is often called positional play, a structured approach to possession in which players occupy specific zones of the pitch to guarantee passing options and to manipulate the opponent's defensive shape. Guardiola's own description of the idea, delivered at a coaching clinic and repeated many times since, is that football is about rationally occupying space, and that the ball moves to the space rather than the other way around. His teams at Barcelona, Bayern Munich, and Manchester City have consistently shown a strong emphasis on short passing, high ball retention, and deliberate patience in possession, often described using the image of a positional grid in which no more than a small number of players occupy the same horizontal or vertical line at once, so that the player in possession always has clear passing options.

A defining feature of Guardiola's approach that recurs across every club he has managed is the use of the goalkeeper and centre backs as active participants in building play, rather than as purely defensive figures whose job is to clear danger. This is reflected in his consistent preference for goalkeepers who are comfortable passing under pressure, and in his willingness to adapt his defensive structure depending on how the opponent presses. At Manchester City in particular, Guardiola became known for inverting his full backs into central positions to create numerical advantages in midfield, and for evolving his attacking structure over time from a front three built around wide players to a system built around a genuine target man once his squad and the tactical landscape

around him changed.

2.2.2 Jurgen Klopp

Jurgen Klopp is most closely associated with a style often called gegenpressing, a German term that translates roughly as counter pressing. The core idea is that the team should try to win the ball back as quickly as possible after losing it, ideally while still high up the pitch and close to the opponent's goal, because a team that has just won the ball in an advanced position is often disorganised and vulnerable to an immediate attack. Klopp himself once summarised his approach by contrasting it with a more patient, possession based style, saying that he preferred football that was loud rather than football that behaved like a quiet orchestral piece.

Across his time at Mainz, Borussia Dortmund, and Liverpool, the consistent thread in Klopp's teams has been an emphasis on transition speed and defensive intensity rather than on patient possession for its own sake. At Liverpool in particular, his teams were built for long periods around advanced, attacking full backs who provided width and crossing threat while a front three combined individual quality with intelligent rotation. In the later years of his Liverpool tenure, this reliance on wide crossing gave way to more central patterns of progression as personnel changed, a reminder that even a manager with a very consistent underlying philosophy adapts the specific mechanisms he uses to express it.

2.2.3 Roberto De Zerbi

Roberto De Zerbi is a more recently prominent figure whose reputation has grown rapidly through his work at Sassuolo, Brighton and Hove Albion, and Olympique de Marseille. His signature mechanism, discussed extensively by tactical analysts, is a deliberate willingness to invite pressure onto his own goalkeeper and centre backs during build up, effectively using the opposition's press against itself. Rather than simply avoiding risk near his own goal, De Zerbi's teams often pause on the ball in deep positions specifically to draw the opponent's press forward, before playing through or around it into the space that press has vacated. This is sometimes described as treating the goalkeeper as an auxiliary defender, splitting the centre backs and adding an extra passing option that the opponent's press line cannot easily account for.

De Zerbi's public profile has risen alongside his results, but his career has also included several difficult and short lived spells at clubs with limited resources, and his time at Marseille ended amid reports of dressing room friction after a period of poor results. This project treats these difficult spells honestly rather than ignoring them, since they matter both for understanding the man and for understanding what data is available about him.

2.3 Why comparing these three managers is genuinely difficult

It would be easy to assume that Guardiola, Klopp, and De Zerbi are simply three different points on some spectrum from patient possession to direct pressing football, and that separating them

statistically should therefore be straightforward. The research conducted for this project suggests the truth is more complicated. Guardiola and De Zerbi, despite very different public reputations, share a great deal in common at the level of underlying mechanism. Both build deliberately through the goalkeeper and centre backs, and both treat possession as an instrument for manipulating the opponent rather than as an end in itself. The place where they are expected to differ most is not in how much they retain the ball, but in the specific sequence of actions they use to do so, something that is much harder to capture with simple statistics than raw possession share.

Klopp, meanwhile, is expected to be easier to separate from De Zerbi specifically, because pressing intensity and transition speed are comparatively well captured by standard statistics, and because his underlying philosophy is explicitly about what happens in moments of transition rather than about sustained patient possession. This expectation, formed early in the project through careful reading of tactical analysis of all three managers, turned out to be an important guide for what the data later showed regarding De Zerbi, though not, as later sections explain, for the specific comparison between Klopp and Guardiola, and this nuance is revisited throughout this report.

2.4 Key technical concepts used in this project

Several technical ideas recur throughout the rest of this report, and it is worth introducing them here in plain terms before they appear in a methodological context.

A feature, in the sense used throughout this project, is simply a number that describes some measurable aspect of a player's or a team's behaviour, such as the number of passes a player completes into the final third of the pitch. A ratio feature expresses one quantity as a share of another, for example the fraction of a player's passes that are progressive, rather than as a raw count. Ratio features tend to be less sensitive to how much a player's team dominates the ball overall, which matters a great deal in this project, as later sections explain.

A z score is a way of expressing how unusual a value is relative to a comparison population, by measuring how many standard deviations it sits above or below the average for that population. Throughout this project, every feature is z scored against the population of players in the same tactical role, so that a defensive midfielder is compared to other defensive midfielders rather than to strikers.

Cosine similarity is a mathematical way of measuring how similar the direction of two lists of numbers is, regardless of their overall size. In this project it is used to compare a player's profile across several tactical dimensions with a manager's profile across the same dimensions, producing a single similarity score. It is important to note, and this point is repeated throughout the report because it matters so much, that cosine similarity is not a probability. A high similarity score means a player's profile points in a similar direction to a manager's profile, not that the player has a certain percentage chance of succeeding under that manager.

Clustering is a family of techniques that group similar items together without being told in advance what the groups should look like. In this project, clustering is used to discover natural groupings of playing styles among thousands of players, independently of the three managers, as a way of

checking whether the tactical dimensions used elsewhere in the project actually capture something meaningful.

Permutation importance is a simple and intuitive way of measuring how much a particular feature contributes to a model's ability to make correct predictions. It works by randomly shuffling that one feature across all the data, breaking its real relationship with the outcome, and then measuring how much the model's accuracy drops. A feature whose shuffling causes a large drop in accuracy is doing real work. A feature whose shuffling causes little or no drop, or even an improvement, is probably contributing noise rather than signal.

2.5 The football knowledge ontology in more detail

Since the football knowledge layer is one of the central ideas of this project, it is worth walking through, in plain terms and before any code or statistics are introduced, how the layer is structured from top to bottom.

At the very bottom sit raw events and raw match statistics, the kind of numbers a data provider publishes directly, such as the number of passes a player attempted in a given match, or how many of those passes reached the final third of the pitch. These numbers on their own do not mean very much to a football person, because they are not normalised for playing time, position, or the strength of the opposition, and because a single statistic rarely tells a complete story by itself.

The next tier up is what this project calls a football concept. A concept takes a small number of related raw statistics and combines them into something a scout or analyst would actually recognise as a single idea. Progressive passing is a good example. Rather than looking at the raw count of progressive passes a player made in a match, which would heavily favour players who simply play more minutes or whose team has more of the ball, the concept is built from the share of a player's total passing attempts that qualify as progressive, combined with the share that reach the final third, the share that reach the penalty area, and the player's overall pass completion rate as a check on quality. Expressing these as shares rather than raw counts turned out, during the project's own evaluation, to matter a great deal, because a share is far less sensitive to how much possession a player's team happens to have overall.

The top tier is what this project calls a tactical dimension, a broader theme built by combining several concepts together using weights chosen deliberately on football grounds rather than purely by statistical convenience. Build up directness, for example, combines the progressive passing concept with a separate possession retention concept and, following the football analyst review described later in this report, a further concept describing how a team's goalkeeper distributes the ball. The specific weights given to each concept inside a dimension were argued explicitly, on football grounds, rather than simply fitted to whatever combination happened to produce the most impressive looking result, because the project's guiding philosophy throughout was that football understanding should shape the model rather than be replaced by it.

Every single concept used in the final system is documented in a dedicated reference maintained

alongside the project's code, recording its precise definition, what it means in football terms, exactly which raw statistics feed into it and how, how it is normalised, and, crucially, its known limitations, including any case where the concept simply cannot be computed for a particular player because the underlying data does not exist. This last point matters more than it might first appear. A system that quietly ignores the difference between a genuinely low score and a score that could not be computed at all risks reporting confident sounding conclusions from an absence of evidence, and one of the more important lessons of this project, discussed at length later, is exactly what happens when that distinction is not respected carefully enough.

With this vocabulary in place, the report can now turn to what has already been done in this field, before describing the work carried out in this project itself.

3. State of the Art

3.1 The broader field of football analytics

Football analytics has grown enormously over the past decade, driven largely by the increasing availability of detailed event and tracking data from providers such as Opta, StatsBomb, and Wyscout. Much of the early academic and industry work in this space focused on evaluating individual actions, for example estimating how much a given pass or shot contributed to a team's chances of scoring. Attention has since broadened to include team level questions, and identifying a team's playing style from data has become one of the most active areas of research in the field.

3.2 A systematic review of playing style identification

One of the most valuable sources consulted for this project is a systematic review by Gu, Wang, Plakias, and Zhang, published in the International Journal of Sports Science and Coaching, which examined twenty five peer reviewed studies on identifying team playing styles using statistical dimensionality reduction and unsupervised machine learning. This review found that principal component analysis, a technique for reducing many correlated statistics into a smaller number of underlying factors, is by far the most common method used in this literature, appearing in around forty four percent of the studies reviewed. It is typically followed by a clustering algorithm such as k means, applied to the factors produced by the dimensionality reduction step, in order to group teams or matches into distinct styles.

The review also highlighted several consistent shortcomings across the field. Very few studies adjust their features for possession share when measuring defensive or pressing behaviour, despite the clear risk that a team with less of the ball will naturally record more defensive actions regardless of its actual pressing intent. Very few studies assess whether their chosen number of clusters is actually the right one using proper validity measures, and many studies work with a single season of data, which cannot capture how a team's style might change under managerial or roster changes. These gaps directly informed decisions made later in this project, particularly

the decision to possession adjust pressing statistics rather than leaving them as raw per match counts.

3.3 A validated taxonomy of playing styles

A particularly important source for this project is a study by Plakias and colleagues, published in the Journal of Physical Education and Sport, which analysed almost six thousand team match observations across eleven European leagues and derived nineteen distinct tactical dimensions describing playing style, each expressed as a scale running between two opposite poles, for example a build up phase that can be more possession based or more direct, and a pressing approach that can be higher or deeper. These nineteen dimensions were validated using discriminant analysis, a statistical technique that checks whether the dimensions actually succeed in telling different groups of teams apart, and the validation results were strong, with correct classification rates between roughly ninety one and one hundred percent across three separate comparisons.

This study is especially relevant to the present project for a reason beyond its methodology. The authors' own clustering results placed Barcelona, Bayern Munich, and Manchester City together in a single style category associated with structured positional possession play, and placed Liverpool in a different category associated with controlled possession combined with rapid counter attacking, alongside Atalanta. Since Barcelona, Bayern Munich, and Manchester City were all managed by Pep Guardiola at different points, and Liverpool was managed by Jurgen Klopp for the period covered by that study, this result offers independent, external evidence that these two managers do leave a detectable signature across different clubs, at least at the level of whole team behaviour observed by an outside research team using its own data and its own methodology. This finding was treated throughout the present project as an important piece of corroborating evidence, separate from anything produced by this project's own model, and it is referenced again in the results and conclusion sections.

3.4 Related systems that go beyond simple statistics

Not every relevant piece of prior work relies on the kind of match statistics used in this project. A notable example is TacticAI, a system developed by Google DeepMind in close collaboration with Liverpool Football Club, which focuses specifically on analysing corner kicks using positional tracking data rather than aggregated statistics. TacticAI represents a football scene as a graph, with players as nodes and their spatial relationships as edges, and uses this richer representation to predict outcomes and suggest tactical adjustments. Crucially, the system was evaluated not just with statistical accuracy measures but through a structured, blinded evaluation involving five football experts from Liverpool Football Club, who were asked to judge the plausibility and usefulness of the system's outputs without knowing which outputs came from the model and which were real recorded outcomes. This evaluation protocol, involving multiple independent experts, held out test cases, and a comparison against a simple baseline, is one of the most rigorous examples of football specific model evaluation available in the public literature, and it served as a template, discussed further in the methodology section, for how this project approached the idea of checking

whether its own outputs made football sense.

The comparison with TacticAI also makes an important point about the limits of the present project. TacticAI works with detailed positional tracking data, which captures exactly where every player is at every moment, and can therefore see team structure directly. The data available for this project, described in detail in the methodology section, consists of aggregated match statistics rather than positional tracking data. This means the present project can see the outcomes of a manager's structural choices, such as how many progressive passes a player completes, but cannot directly see the structure itself, such as the shape of the team or the spacing between players. This distinction turned out to be one of the most important limiting factors discovered during the project's own evaluation, and it is discussed at length in the results and conclusion sections.

3.5 Other relevant work

Several other studies consulted for this project contributed specific methodological ideas. A study by Zhong and colleagues, published in Biology of Sport, used an autoencoder, a type of neural network capable of capturing nonlinear relationships, to classify playing styles in the Champions League and to predict match outcomes from those styles, explicitly arguing that linear techniques such as principal component analysis can fail to capture some of the more complex relationships present in football data. That same study is also notable for demonstrating that individual statistics can have opposite relationships with success depending on the playing style being examined, for example showing that crossing frequency was associated with winning for possession dominant teams but associated with losing for teams that relied more on direct counter attacking. This finding was treated as an important piece of supporting evidence for a principle followed throughout the present project, that a single statistic rarely has a fixed, universal footballing meaning independent of the broader tactical context it appears in, which is precisely why raw statistics were never used directly and were always passed through the football concept and dimension layers described earlier in this report.

A study of clustering football game situations through deep representation learning, by Tang, Wang, and Zhang, took a different approach again, learning representations of specific in game situations directly from spatial and event data rather than starting from a predefined list of statistics, and using those learned representations to group situations with similar tactical character. This work was read carefully as a reminder that the concept based approach followed in this project, while chosen deliberately for its interpretability, is not the only way to approach the underlying problem, and that a system willing to give up some interpretability in exchange for a richer, learned representation can capture patterns that a hand built concept layer might miss entirely.

A masters thesis by Bolin and Vesterlind at Uppsala University identified offensive and defensive playing styles from possession chains and explored practical applications for club management, including a specific, if limited, attempt to project an individual player onto a different club's style using a single summary metric, offering a directly comparable example of an academic project attempting to translate statistical style identification into something a club could actually use, and

one of the very few sources found during this project's literature review that even attempts the player to system projection idea this project focuses on, albeit in a considerably simpler form. A capstone project by Baattite at Al Akhawayn University applied exploratory factor analysis to a set of thirty eight key performance indicators and produced named, interpretable tactical styles, offering a template for how factor loadings can be translated into football language rather than left as abstract statistical output, an approach that directly informed how this project chose to translate its own tactical dimensions into plain language explanations inside the final application.

Across almost every one of these studies, and this point is worth stating plainly because it shaped several concrete decisions in this project, remarkably few explicitly adjust defensive or pressing statistics for how much possession the team being measured actually had. Of the studies reviewed in the systematic review discussed above, only a small minority report doing this at all. This gap in the wider literature is precisely why possession adjustment of pressing statistics, described in the methodology section of this report, was treated as an important improvement worth implementing directly, rather than as an optional refinement.

3.6 What has not been done, and where this project fits

Taken together, the existing literature demonstrates convincingly that identifying playing styles from data is a mature and active area of research, with several validated methodologies and taxonomies available. What the literature does not do, at least among the sources consulted for this project, is model playing philosophy at the level of an individual manager, as distinct from a club or a single team season. Every study reviewed treats the team, or occasionally an individual match, as the unit of analysis. None of them ask whether a manager's style can be tracked as he moves between clubs, none of them build an explicit compatibility score between an individual player and a specific manager's philosophy, and none of them make the mapping from raw statistics to interpretable tactical concepts as explicit and auditable as this project attempts to make it.

It is also worth being honest about a limitation of the literature review itself. While preparing the review, a reference was identified, a paper by Cintia and Pappalardo sometimes referred to by the name Coach2vec, which appears from its title and abstract to model coaching style directly using an embedding approach. This paper was not part of the set of documents supplied for this project and was not read in full, but its existence is noted here explicitly, because any claim that this project is the first to model manager level style specifically would need to be checked against that paper before being made with confidence. This project therefore makes a more modest and defensible claim, that it demonstrates a knowledge informed approach to manager level compatibility modelling with a strong emphasis on explainability, built and evaluated with a documented level of rigour, rather than claiming to be the absolute first attempt at the underlying idea.

4. Methodology and Work Made

This section describes everything that was actually built during the project, in roughly the order it was built, following the CRISP DM methodology that was adopted as the overall framework, covering business understanding, data understanding, data preparation, modelling, evaluation, and deployment. The section is deliberately detailed, because the process of investigation, discovery, and correction that took place is as important to the project's credibility as any single result.

4.1 Business understanding and the non negotiable rule

Before any data was collected, one rule was established as non negotiable and shaped every decision that followed. A player's statistics may only be attributed to a manager for the exact period that manager was actually in charge of the relevant team, checked at the level of individual matches rather than whole seasons. This rule exists because managerial changes very often happen in the middle of a season, and a naive approach that simply assigned every match in a season to whichever manager happened to be in charge at some point during it would silently mix together data from completely different tactical regimes. Two of the richest, most heavily documented seasons available in the public data used for this project, Liverpool's 2015 and 2016 season and Brighton's 2022 and 2023 season, both include a mid season managerial change, which makes this rule anything but a theoretical concern.

4.2 Building a verified manager tenure record

The first substantial piece of work was constructing a verified record of every club each of the three managers has led, with start and end dates checked against at least two independent sources for every entry. This record covers twenty one distinct tenures across the three managers, including early career spells at lower profile clubs that turned out to be important for understanding data availability later. Every tenure was graded according to how safely it could be used for statistical attribution. Tenures that covered a whole season with no managerial change during that period were graded as safe. Tenures that began or ended partway through a season were flagged as requiring match level splitting rather than a simple season level assignment. A small number of tenures, such as De Zerbi's short and difficult spell at Palermo, involved so many managerial changes within a single season, in that case five different managers in one campaign, that they were flagged as unsafe to use at all without extremely careful handling.

This exercise also surfaced and corrected several common misconceptions found in less careful secondary sources. For example, several biography style web pages claim that De Zerbi won promotion with two of his early clubs, Darfo Boario and Foggia. Checking primary sources, including contemporary Italian sports press coverage, showed that both of these promotion claims are false. De Zerbi's Foggia side lost the promotion play off final on penalties, and the actual promotion of that club came the following season under a different manager. This kind of careful verification, while time consuming, matters a great deal for a project whose entire premise depends on correctly attributing data to the right manager.

A handful of other tenures deserved particular care, and are worth describing individually because

each illustrates a different kind of trap that a less careful project might fall into. De Zerbi's short spell at Palermo in the 2016 and 2017 season is perhaps the most extreme example encountered during this project. He was appointed after the club's previous manager left following only two matches, and he himself was dismissed after just thirteen matches and a single win, after which the club went on to use three further different managers before the end of that same season, meaning five separate managers led Palermo within one campaign. Any attempt to attribute a whole season of Palermo statistics to a single manager would have been meaningless, and this tenure was ultimately treated as too fragmented to use for anything beyond the historical record itself.

Klopp's arrival at Liverpool in October 2015 is a different kind of trap, because it involves two dates that are very easy to confuse if a source is not read carefully. Liverpool's previous manager was dismissed after a Merseyside derby early in the season, and Klopp was announced as the new manager a few days later, during an international break, before taking his first match in charge against Tottenham Hotspur several days after that announcement. A careless attribution based on the announcement date alone would have wrongly assigned more than a week of matches, including at least one Liverpool fixture, to the wrong manager. The verified tenure record built for this project deliberately distinguishes between the date an appointment is announced and the date of a manager's actual first match in charge, and uses only the latter for any statistical attribution.

A related trap appeared with Guardiola's move to Manchester City. His appointment was publicly announced several months before he actually took charge, since the previous manager remained in his post for the remainder of that season. Several otherwise reliable looking summary sources list the announcement date as though it were the date his tenure began, which would have wrongly attributed a large portion of a Premier League season to Guardiola before he had ever taken charge of a single match. This is exactly the kind of subtle error that the project's insistence on checking every date against at least two independent sources, and on distinguishing announcement dates from effective dates, was specifically designed to catch.

4.3 Auditing what data is actually available

With a verified tenure record in place, the next task was to determine, through direct investigation rather than assumption, which public data sources actually cover these three managers with sufficient statistical detail. Two main sources were investigated in depth.

The first source, StatsBomb Open Data, is a public collection of detailed event level football data released by StatsBomb, a leading data provider. This source proved extremely useful for one specific period, Guardiola's four seasons at Barcelona, for which it provides rich, event by event detail covering more than one hundred team matches. For every other relevant period, however, this source proved far too limited to serve as a primary data source. Klopp appears in only thirty three team matches across his entire career within this dataset, and De Zerbi does not appear at all, since none of his clubs or seasons happen to fall within the specific competitions and seasons this public dataset happens to cover.

The second source, a public mirror of detailed match statistics originally collected by Opta and made available through FBref, proved to be the primary data source for the project. Direct inspection of this source, downloading and checking the actual contents rather than trusting documentation, confirmed that it covers the top five European leagues, meaning the English Premier League, the Spanish La Liga, the Italian Serie A, the German Bundesliga, and the French Ligue 1, with match by match detail from the 2017 and 2018 season onward. This determined the practical scope of the project. Guardiola's earlier spells at Barcelona and Bayern Munich, and Klopp's entire time at Borussia Dortmund, fall before this window and are therefore outside what could be modelled quantitatively, a limitation discussed candidly in the results and conclusion sections rather than hidden.

Given the explicit importance the project placed on not simply accepting a convenient data limitation without checking whether it could be overcome, an active search was carried out for alternative sources that might extend coverage back to these earlier periods. This search included checking whether FBref itself had ever published detailed statistics for these earlier seasons under a different access path, checking a separate public repository of basic match results going back decades, and checking a well known Kaggle dataset covering European football from 2008 to 2016. Each of these alternatives was tested directly rather than dismissed on assumption. The FBref check confirmed, by looking at whether expected goals figures existed for matches in that period, that detailed statistics simply do not exist on the source website itself for that era, meaning this is a genuine gap in what was ever recorded publicly, not merely a gap in this project's chosen mirror of that data. The basic results repository was confirmed to contain only scores and dates, with no passing or defensive detail at all. The Kaggle dataset was found, on inspection of its actual schema, to contain video game style player ratings rather than measured match performance, which would not have been usable even if it had been fully downloaded. This search is documented in detail because it reflects an important principle followed throughout the project, that a data limitation should only be accepted once it has genuinely been tested, not merely assumed.

One further correction emerged from this careful checking process. An initial pass at scoping the project's data had assumed that De Zerbi's short spell at Benevento, cut short partway through the 2017 and 2018 season, had no usable detailed match data because it fell in the same period that lacked coverage for Italian football generally. A more careful check found that this specific season was in fact covered, and the correction was made to include it, flagged as a lower confidence addition given the well documented difficulty of that particular Benevento squad.

A separate and important data source, discovered part way through the project when it became clear that the match level data described above had a specific gap, is a season level aggregated version of the same underlying statistics, also mirrored publicly. Checking this source directly revealed that, unlike the match level data, it has complete coverage across all five leagues with no gap for Italian or French football, because the gap in the match level data turned out to be specific to how that particular version of the data was compiled, not a reflection of what FBref itself ever published. This discovery, and how it was used to substantially improve the project's results, is described in detail later in this section.

4.4 Checking availability for South American leagues

Partway through the project, a specific request was made to extend the player database used by the final application to include top South American leagues alongside the European ones, so that a user could search for and evaluate players from those competitions as well. This request was investigated directly and thoroughly rather than assumed to be either easy or impossible.

The primary match level data source used elsewhere in the project, on direct inspection of every league code it makes available, was found to cover only six countries, the five European leagues already mentioned plus a top American league, with no South American competition included at all. A separate release from the same broader data project was found to include some raw event level data for the Brazilian top division, but on inspection this data consisted only of goals, cards, and similar discrete events, with no passing, possession, or defensive statistics comparable to what the rest of the project relies on. A widely used alternative source, Understat, was checked directly and confirmed to cover only the same five European leagues already in use, with no South American coverage whatsoever. The StatsBomb Open Data source, already investigated for other reasons, was confirmed not to include any current South American league data either.

Given this direct, exhaustive check across every accessible source, the conclusion reached was that no source available to this project provides South American league statistics at a level of detail comparable to what is available for the top five European leagues. Rather than either silently ignoring this limitation or spending further time on a search that had already been shown to be unproductive, the decision was made to scope the player database to the top five European leagues only, and to state this limitation clearly and visibly, including inside the final application itself, so that a user is never left wondering why a South American player cannot be found.

4.5 The football knowledge layer

With data availability understood, the project moved to what is arguably its most distinctive methodological contribution, an explicit football knowledge layer that sits between raw statistics and any modelling step. The guiding principle behind this layer is that a system should never take dozens of raw statistics and feed them directly into a model without passing through an intermediate stage that a football person would recognise and could critique.

The knowledge layer was built in three tiers. At the lowest tier sit raw match statistics, such as the number of passes a player completed into the final third of the pitch during a given match. These raw statistics are combined into what this project calls football concepts, each representing a single, named idea that a scout or analyst would recognise, such as progressive passing, pressing intensity, or box presence. Each concept is built from a small, carefully chosen set of underlying statistics, and every concept in the final system is documented with its precise definition, its football meaning, exactly how it is calculated, how it is normalised, and its known limitations, in a dedicated data dictionary maintained alongside the code.

At the top tier, these concepts are combined into a smaller number of tactical dimensions, each

representing a broader theme such as build up directness, chance creation, or individual defending. The specific set of tactical dimensions used in this project was not invented from nothing. It was anchored explicitly on the validated nineteen dimension taxonomy from the Plakias study described in the state of the art section, adapted and extended using both the wider academic literature and the project's own detailed research into the three managers' known tactical characteristics. Where a dimension or concept in the final system reflects something identified in the project's own manager research rather than something explicitly named in the academic literature, such as goalkeeper distribution style, this is stated openly rather than presented as if it came from a validated external source.

The combination of raw statistics into concepts, and concepts into dimensions, follows a hybrid approach agreed early in the project. Principal component analysis is used as a check, confirming that the specific statistics grouped together under a given concept actually do move together statistically, which validates that the grouping reflects something real in the data rather than an arbitrary choice. The actual weights used to combine concepts into dimensions, however, are set explicitly based on football judgement rather than purely statistically derived, because the project's guiding philosophy throughout was that football knowledge should shape the model, not simply be replaced by whatever a purely data driven method happens to produce.

4.6 A worked example of how a concept becomes an explanation

It is easier to appreciate why the football knowledge layer matters by walking through a single concrete example from beginning to end, following the pressing intensity concept, since it turned out to be the most important dimension in the project's final results.

The starting point is a small set of raw defensive statistics recorded for every player in every match, including the number of tackles made in the attacking and middle thirds of the pitch, the total number of tackles made anywhere on the pitch, the number of interceptions, and the number of loose balls recovered. None of these numbers on their own says anything reliable about a manager's pressing philosophy, because a player on a team that spends very little time in possession will naturally accumulate more of these actions simply through greater exposure to defending, regardless of how aggressively his team actually presses.

The concept layer first converts each of these raw counts into a rate per ninety minutes, so that a player who only came on as a substitute is compared fairly with one who played a full match. Following the football analyst review described later in this section, each of these per ninety minute rates is then further adjusted using real team level possession data, scaling a team's raw rate upward if its opponents tended to have more of the ball against it, and scaling it downward if its opponents tended to have less, so that the final figure reflects pressing intent rather than simply how much defending a team was forced to do. These adjusted rates are then averaged together into a single pressing intensity concept score for that player in that match.

The dimension layer then combines this concept, along with everything else computed for that player, into the player's overall tactical profile, and this profile is compared against each manager's

own profile using cosine similarity. When the final application displays a result to a user, it does not stop at the similarity number itself. It decomposes that number back down into its contributing parts, and if pressing intensity turns out to be one of the dimensions pulling most strongly toward a match with a particular manager, the application reports this directly, stating the player's pressing intensity score relative to others in his position and naming the specific contribution that dimension made to the overall similarity. This is what allows the system to produce a sentence a scout could actually use, rather than a number that would need a data scientist standing next to it to be interpreted at all.

4.7 Building the manager attributed dataset

The next stage of work combined the verified tenure record with the audited data sources to build an actual dataset in which every player match observation is correctly tagged with the manager in charge at that exact moment. This required writing dedicated code that filters the raw match statistics down to the exact date ranges established for each manager, handling every mid season change identified earlier by splitting a club's data at the precise date of the managerial change rather than at the boundary of a season.

Once this dataset was built, a dedicated integrity check was run before any further work proceeded, checking that every row fell inside its correct manager assigned date range, that no duplicate observations existed, and that the resulting match counts made sense against what was already known from the verified tenure record. This check passed cleanly, and in doing so also produced an unplanned but valuable confirmation of the tenure record's accuracy. For example, Brighton's 2022 and 2023 season produced exactly thirty two matches attributed to De Zerbi, which matches precisely what would be expected from a thirty eight match season minus the six matches played before his arrival, an independent cross check that the whole pipeline was working as intended.

4.8 A first version of the compatibility model, and the problem it revealed

With a verified, integrity checked dataset in place, a first working version of the full pipeline was built. Concept and dimension scores were computed for every player, manager profiles were built as the average profile of every player who played under that manager, weighted by minutes played, and a compatibility score was computed between any player and any manager as the cosine similarity between their two profiles. Every score was made explainable by decomposing it into the contribution of each individual tactical dimension, so that the system could always report which specific characteristics were driving a given result, rather than presenting a bare number.

An important validity check was built alongside this model, directly addressing one of the most serious risks identified from the outset, that a model might appear to detect a manager's philosophy while actually just detecting which club a player happened to play for. Since De Zerbi is the only one of the three managers whose available data spans several different clubs, comparing his own club by club profiles to each other, and to Guardiola's and Klopp's single club profiles, provides a genuine test of whether the model captures something that survives a change of club and league,

or whether it is simply relearning club identity.

The result of this first working version was striking, and initially looked like a serious problem. Two of De Zerbi's clubs, Sassuolo and Benevento, appeared statistically almost the exact opposite of every other club in the dataset, while every other club, regardless of which manager it belonged to, clustered tightly together. Rather than accepting this as evidence about football, the result was investigated, because a pattern this clean and this extreme looked more like an artefact than a real finding, and this instinct turned out to be correct.

4.9 Two genuine bugs, found and fixed rather than reported as findings

The investigation uncovered two distinct, serious problems, both of which are documented here in detail because finding and correcting them is arguably the single most important demonstration of methodological rigour in the entire project.

The first problem was a coding error in how missing data was handled. Wherever a particular statistic was simply not available for a given league and season, the code was silently treating that missing value as if it were a real recorded zero, effectively telling the model that a player had done absolutely none of a given action, rather than correctly recording that no information existed at all. This is a serious error because it makes a genuine data gap look exactly like an extreme, real behavioural signal.

The second problem, uncovered while trying to understand why that coding error had such a large effect specifically on Italian football, was a previously undetected gap in the underlying data source itself. Detailed passing, possession, and goalkeeper statistics turned out to be completely absent from the match level data source for Italian and French football before the 2023 and 2024 season, even though matches themselves were present in the data with correct scores and player names. This meant that Sassuolo and Benevento, both Italian clubs during the relevant period, appeared to have wildly unusual profiles not because De Zerbi's teams actually played that differently, but because five out of seven tactical dimensions were being computed from data that simply did not exist for those specific rows, combined with the first bug that was turning that absence into a false, extreme signal.

Both problems were corrected. The handling of missing data was rewritten throughout the entire pipeline so that a genuine gap always remains recorded as unknown and is properly excluded from any average it would otherwise have distorted, rather than ever being silently treated as zero. Every comparison in the system was also updated to record, transparently, how many of the seven tactical dimensions were actually available for a given comparison, so that a comparison based on only one or two dimensions is clearly flagged as low confidence rather than treated with the same weight as a comparison using all seven.

Once these corrections were made, the extreme, suspicious pattern disappeared, and a more modest, more believable picture emerged, in which the model's ability to tell the three managers apart was only marginally better than random guessing. This was, in its own way, a disappointing result, but it was also a far more trustworthy one than the original, dramatic seeming pattern, precisely

because it had survived a genuine attempt to break it rather than being accepted on first sight.

4.10 Recovering lost data using the season level source

Having fixed the coding errors, attention turned to the underlying data gap itself. Since a separate, season level version of the same broadly sourced statistics had been identified earlier as having full coverage where the match level version did not, dedicated code was written to bring this season level data into the pipeline specifically for the periods affected by the gap. Sassuolo's entire three season spell under De Zerbi, which had previously contributed almost nothing to five of the seven tactical dimensions, could now be computed from genuine, complete statistics rather than missing data.

This correction also revealed a further, welcome discovery. The season level data source, for seasons up to 2021 and 2022, still contains a specific measure of pressing actions collected under an earlier data provider, a measure that had been dropped from later seasons and was never present in the match level source at all. Since pressing intensity had already emerged as the single most promising tactical dimension in the project's results, described in detail in the next section, this additional, more direct measure of pressing was incorporated as a further input to that dimension wherever it was available, again with the missing data handling described above ensuring it degraded gracefully rather than breaking anything for the periods where it was not.

Bringing this richer data into the pipeline produced a real, measurable improvement in the model's ability to tell the three managers apart, described fully in the results section, and it substantially narrowed, though did not fully close, the gap between De Zerbi's own clubs that had originally looked so suspicious.

4.11 A football analyst review of the whole pipeline

At a later stage of the project, once a full working pipeline existed, a deliberate step was taken to review the entire approach from the perspective of a football analyst rather than purely as a technical exercise, asking specifically what a person with real tactical expertise would criticise about the system as built. This review identified several genuine weaknesses, three of which were judged both important and practically fixable with the data already available, and were implemented as a direct result of this review.

The first weakness identified was that players were being grouped into only four very broad positional categories, goalkeeper, defender, midfielder, and forward, for the purpose of comparing each player against others in a similar role. A defensive midfielder, a central midfielder, and an attacking midfielder have very different natural baselines on almost every tactical dimension, and lumping them together adds noise that could easily be mistaken for a lack of managerial signal when it is really just role variation. This was corrected by building a much more detailed eight category role system, using specific positional codes already present in the match level data, such as centre back, full back, defensive midfielder, central midfielder, attacking midfielder, winger, and striker, falling back to a simpler zone based estimate only for the smaller amount of data that lacked

this detail.

The second weakness identified was that pressing intensity, despite being the most promising dimension in the project's results, was still being measured as a simple rate of defensive actions per ninety minutes, with no adjustment for how much of the game a team's opponents actually spent in possession. A team that has less of the ball overall will naturally record more defensive actions simply because it spends more time facing an opponent in possession, regardless of how aggressively it presses. This was corrected by building a possession adjustment using real team level possession statistics, scaling each team's raw defensive rate by how much possession its opponents typically had, so that the resulting measure more fairly reflects pressing intent rather than pressing opportunity.

The third weakness identified was that a specific data source, detailed goalkeeper distribution statistics such as average pass length and how often a goalkeeper plays long, launched passes rather than short ones, had already been collected as part of the pipeline but was never actually used by any part of the model, despite the project's own research into the three managers explicitly identifying goalkeeper build up behaviour as one of the clearest ways of distinguishing Guardiola and De Zerbi's approach from a more direct style. A new tactical concept was built specifically to capture this, and incorporated into the broader build up dimension.

Implementing these three changes required one further, less visible but equally important correction. The way tactical dimensions were combined from their underlying concepts had been written in a way that assumed every concept would always be available for every player. Since the new goalkeeper distribution concept is, by its very nature, only ever available for goalkeepers, this assumption would have silently broken the build up dimension for every single outfield player the moment the new concept was added. This was caught and fixed before it could cause a problem, by rewriting the combination step so that it correctly adapts to whichever concepts are actually available for a given player, rather than assuming all of them always are.

A further complication was discovered and corrected during the implementation of the new positional system itself. An initial version of the eight category role classifier produced a badly broken result, placing the overwhelming majority of all players into a single category, because it had incorrectly assumed that the match level data used the same broad positional labels as the season level data, when in fact the match level source uses a much more specific and different set of position codes. This was identified immediately by inspecting the actual distribution of results rather than assuming the code was correct, and was corrected before any further analysis relied on it.

Together, these three targeted improvements, informed by football expertise rather than by generic data science practice alone, produced the single largest improvement in the project's results of any individual change made during the project, described fully in the next section.

4.12 Clustering and evaluation methodology

Alongside the manager focused compatibility model, a separate clustering analysis was carried out across the full population of over eleven thousand player seasons collected from the top five

European leagues, entirely independent of the three managers, in order to check whether the tactical dimensions built for this project capture something meaningful on their own terms. The k means clustering algorithm was used, chosen because it is well established, easy to interpret, and consistent with the modal approach identified in the academic literature reviewed earlier in this report.

Rather than simply picking an arbitrary number of clusters, several candidate values were tested and compared using three separate validity measures, the silhouette score, the Davies Bouldin index, and the Calinski Harabasz index, each of which evaluates cluster quality from a slightly different angle. The results consistently suggested that football playing styles do not separate into sharp, clearly distinct groups, a conclusion that is itself informative rather than disappointing, since it matches the widely held view among football analysts that playing style exists on a continuum rather than in discrete categories. A final set of five clusters was chosen as a practical, interpretable compromise, each automatically labelled using its two strongest tactical dimensions, producing recognisable groupings such as players strong in defending and pressing, and players strong in build up and progressive passing.

The main compatibility model itself was evaluated using a deliberately strict, adversarial standard, checking what fraction of player seasons have their single best matching manager, according to the model, actually be the manager they played under in real life, restricted to comparisons where enough of the seven tactical dimensions were genuinely available to make the comparison meaningful. This is a demanding test, because even a genuinely accurate model of tactical philosophy would not achieve perfect accuracy on this measure, since players do not get freely assigned to clubs purely on the basis of tactical fit. Contracts, transfer fees, and existing squad composition all influence who actually ends up playing for whom. This limitation of the evaluation approach is discussed further in the conclusion section, but the measure was kept as the primary evaluation criterion throughout the project because it is concrete, checkable, and resistant to being talked around.

As a further explainability check, since the standard technique known as SHAP requires a trained predictive model and the compatibility score in this project is a similarity calculation rather than a trained classifier, permutation importance was used instead, following the same underlying logic, to measure how much each individual tactical dimension actually contributes to the model's ability to correctly match players to their real managers.

4.13 Building the interactive application

The final major piece of work was building an interactive application, using the Streamlit framework, that makes the entire pipeline usable by someone with no technical background at all. The application allows a user to search for any player from the top five European leagues by first choosing a season, then a league, then a specific club, and finally a specific player from that club's roster for that season, mirroring the natural way a scout or analyst would actually think about finding a player.

Once a player is selected, the application displays that player's compatibility with each of the three managers, expressed both as a raw similarity score and as a rescaled index out of one hundred for easier reading, always accompanied by an explicit reminder that this is a similarity measure rather than a probability. For the best matching manager, the application shows a plain language explanation, listing the specific tactical characteristics that contributed most positively to the match, and separately listing any characteristics that worked against it. The application also shows a radar chart comparing the selected player's profile directly against any or all of the three managers, shows which of the five data driven playing style clusters the player falls into, shows where that player sits visually among the entire population of over eleven thousand players using a two dimensional projection of their tactical profile, lists the players statistically most similar to the one selected, and allows a direct side by side comparison between any two players.

Two further refinements were added in direct response to specific feedback during the project. Photographs of the selected player and of each manager are retrieved automatically from Wikipedia, using a lookup that is deliberately conservative, falling back to a simple placeholder image rather than risking showing an incorrect photograph for a player who shares a name with someone else or who does not have a suitable public photograph available. Fixing this feature also required discovering and correcting a specific technical obstacle, since Wikipedia's own systems require a properly identified request before they will return any data at all, something that was not obvious until the very first attempt failed and had to be diagnosed. The application's content was also reorganised into separate tabs, one for manager fit and explanation, one for the tactical profile comparison, one for the playing style cluster view, and one for player comparison tools, so that the amount of information on screen at any one time remains manageable.

Throughout the application, and this was treated as a firm requirement rather than an afterthought, the project's own honestly evaluated limitations are displayed prominently and are not hidden behind a separate technical document that a casual user might never open. A warning describing the model's actual, measured accuracy, and the specific reasons that accuracy is not higher, appears directly on the main screen of the application every time it is used.

4.14 Documentation and reproducibility

A project of this scope, built over a short and intense period of time and involving repeated corrections and improvements, is only as trustworthy as the record it leaves behind of how it actually got to its final state. For this reason, a deliberate and continuous effort was made throughout the project to document decisions as they were made, rather than reconstructing that documentation afterward from memory.

Every material methodological choice, from the decision to treat two different data sources as never directly comparable, to the decision to rewrite how missing data was handled, to the decision to adjust pressing statistics for possession, was recorded in a running decision log at the moment it was made, together with the alternatives that were considered, the evidence that supported the final choice, and the practical consequences that followed from it. This log now contains seven

major entries and serves as the single clearest account of why the system looks the way it does, rather than some other way.

Alongside the decision log, a detailed data dictionary documents every single engineered concept and tactical dimension used anywhere in the final system, following a consistent format that records what each one measures, what it means in football terms, exactly how it is calculated from raw statistics, how it is normalised, and any known limitation, including cases where a concept cannot be computed at all for a particular manager because of a data gap described earlier in this section. This dictionary was kept in step with the actual code throughout the project, including being revised twice as concepts were added or changed, so that at every point in the project it accurately describes what the system does rather than what it was originally intended to do.

The entire analytical pipeline itself was built as a sequence of executable notebooks, organised to follow the stages of the CRISP DM methodology that structures this whole project, covering data collection and manager attribution, feature engineering, player and manager compatibility, season level data integration, and clustering together with visualisation. Every one of these notebooks runs cleanly from start to finish on demand, producing the same results each time, which matters a great deal for a project whose central claim rests on specific, checkable numbers rather than on a single figure asserted without a visible path back to the data that produced it. All of the reusable logic behind these notebooks, including the data loading, the manager attribution rules, the feature engineering definitions, and the compatibility scoring itself, was written as a proper, shared code library rather than being duplicated inside each notebook, so that a fix made in one place, such as the correction to how missing data is handled, automatically applies everywhere that logic is used rather than needing to be repeated and potentially forgotten in several separate locations.

5. Results

5.1 Manager profiles

The final manager profiles, built from every available player observation correctly attributed to each manager and expressed across the seven tactical dimensions used throughout the project, show a picture that is broadly consistent with, but more nuanced than, and in one specific respect different from, the popular narrative around these three managers. Guardiola's Manchester City profile shows the strongest build up and progressive passing scores of the three, consistent with his reputation as the leading exponent of structured possession football. Klopp's Liverpool profile sits close behind Guardiola on build up and progression, which is unsurprising, but on pressing intensity, once that measure was properly possession adjusted, it is actually Guardiola's profile that comes out marginally higher, not Klopp's, which is the opposite of the expectation formed early in the project through careful reading of tactical analysis. The most likely explanation, discussed further in the conclusion, is that Manchester City's possession dominance is so extreme within this sample that the possession adjustment, which corrects for an opponent's average share of the ball rather than for squad quality in general, ends up scaling City's adjusted rate upward by more than

a purely managerial reading of pressing intent would justify. This is best read not as a failure but as a concrete example of the residual squad strength confound this report describes throughout. What does hold up exactly as expected is De Zerbi's position relative to both other managers. De Zerbi's combined profile across his four included clubs sits below both of the other two managers on most attacking dimensions, and clearly below both on pressing intensity, a result that on its own might look surprising given his reputation as an attacking, expansive coach, but which is in fact strongly consistent with the project's own qualitative research into his tactics, which identified his signature mechanism as something that happens while his own team has the ball, drawing the opponent's press forward and playing through it, rather than something that happens after his team loses the ball. A manager whose central idea operates in possession rather than out of it would be expected to show exactly this kind of profile, and finding that the data agrees with this specific, previously stated expectation for De Zerbi, even where it did not agree for the specific ordering between Guardiola and Klopp, is one of the more instructive results the project produced.

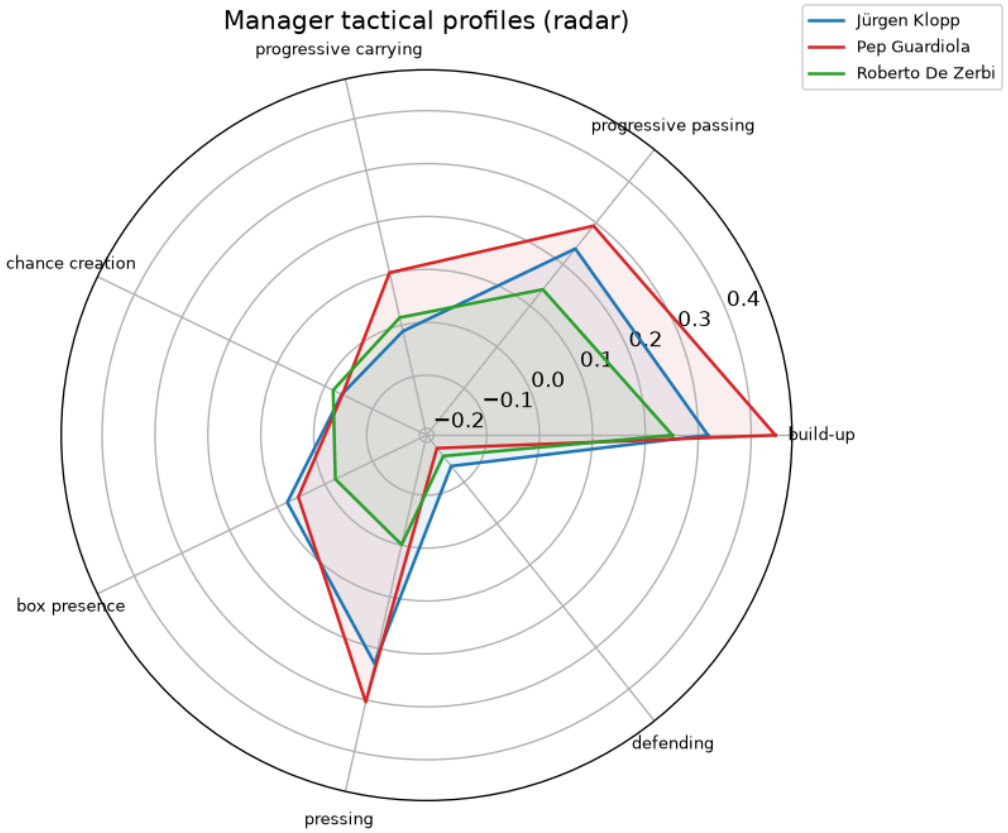


Figure 1: The seven tactical dimension scores for Guardiola, Klopp, and De Zerbi, each measured against the same league wide position baseline. The three profiles sit closer together than the popular narrative around these managers might suggest, which is itself an important finding discussed throughout this section.

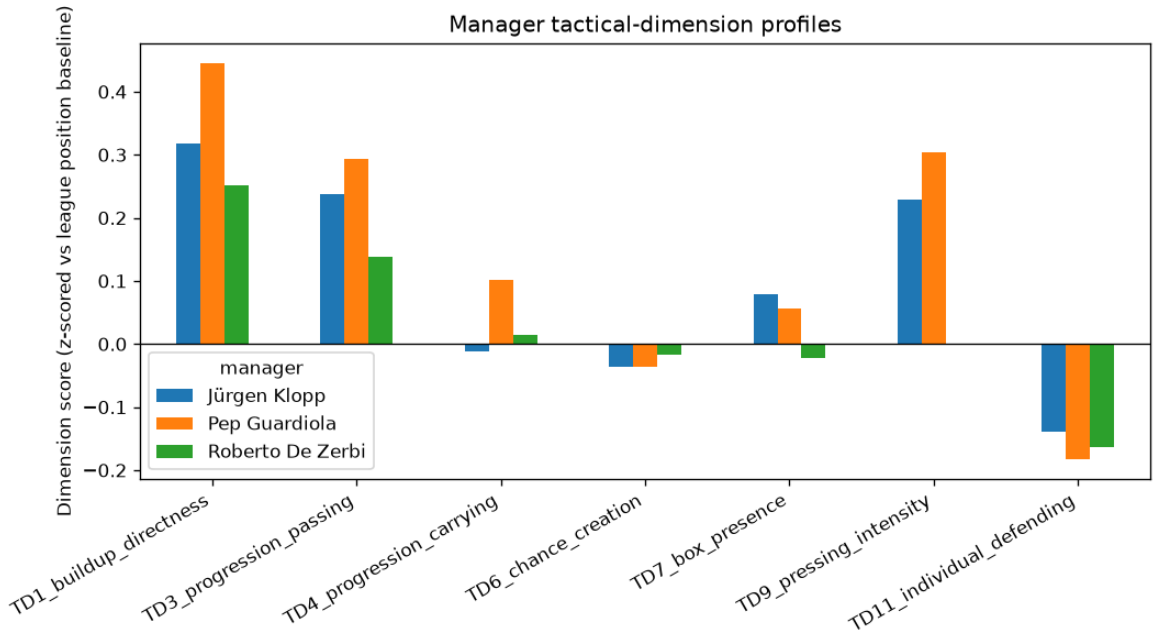


Figure 2: The same three manager profiles shown as a grouped bar chart, making it easier to compare the managers dimension by dimension.

5.2 The evolution of the model's accuracy

The single most important quantitative result of the project is not a single number, but the sequence of numbers produced across three distinct stages of the work, because that sequence tells an honest story about how the project actually progressed.

In the first working version of the model, before the two coding and data errors described in the methodology section were found and corrected, the model appeared to separate the three managers with a striking, almost suspiciously clean pattern. Once those errors were corrected, the model's accuracy at matching a player to the manager he actually played under, measured only on comparisons where enough data was genuinely available, fell to just under thirty three percent, statistically no better than simply guessing among the three managers at random. This was a disappointing but far more trustworthy result than the original pattern, and it was reported honestly rather than quietly discarded.

Recovering the lost Italian and French data using the season level source, described in the methodology section, raised this accuracy to just under thirty eight percent, a modest but genuine improvement. The subsequent football analyst driven improvements, namely the more detailed positional grouping, the possession adjustment applied to pressing statistics, and the new goalkeeper distribution feature, raised the accuracy further to just over forty two percent. This final figure represents a real, meaningful improvement over random guessing, roughly nine percentage points above the one in three baseline that would be expected from chance alone, though it remains well short of anything that could be called a decisive or highly reliable result.

The accuracy progression, stage by stage

- Chance baseline: about 33 percent, since there are three managers to choose from
- First working model, before correction: an extreme, suspicious pattern that turned out to be a data error, not a real result
- After fixing the two genuine bugs: 32.8 percent, statistically no better than chance
- After recovering the missing Italian and French data: 37.8 percent
- After the football analyst driven improvements: 42.1 percent, the project's final, reported result

5.3 Is this result statistically significant

The 42.1 percent figure above is a single point estimate over 561 player seasons. On its own, a point estimate does not say how much of that number could plausibly be noise from this particular sample, nor how unlikely a rate this high would be if the model's decisions actually had no real relationship to the manager a player played under. Two further checks close that gap, both computed directly from the already saved compatibility results, with no new data or model changes required.

The first check is a bootstrap confidence interval. Resampling the five hundred and sixty one player seasons with replacement five thousand times and recomputing the match rate each time gives a ninety five percent confidence interval of thirty seven point eight percent to forty six point two percent, with a median of forty two point one percent. The second check is a permutation test. Shuffling the true manager label among those same rows five thousand times, while leaving the model's own best fit decision untouched, and recomputing the match rate each time, builds an empirical picture of how often a decision this good would arise if there were no real association at all, given this particular sample's actual mix of managers. That empirical chance baseline turns out to be thirty four percent, close to but not exactly the naive one in three figure used elsewhere in this report, because De Zerbi, Klopp and Guardiola are not represented in exactly equal numbers in this sample. Under that empirical baseline, the observed forty two point one percent was matched or exceeded in only one of the five thousand permutations, giving a p value of zero point zero zero zero two.

Two questions, two different answers

- Is this better than chance at all? Yes, confidently. The permutation test gives a p value of 0.0002, meaning the model's best fit decisions line up with the true manager far more often than would be expected if there were no real relationship between them.
- How precisely is the 42.1 percent figure known? Not very precisely. The bootstrap confidence interval spans from 37.8 percent to 46.2 percent, nearly nine percentage points end to end, so the point estimate should be read as somewhere in the high thirties to the mid forties, not as an exact number.

These two results should be read together rather than either one alone. A report that only quoted the point estimate would risk implying more precision than five hundred and sixty one observations

can actually support. A report that only quoted the significance test would risk implying the exact number matters more than it does. Taken together, they say the project's central quantitative claim is real and unlikely to be a fluke of this particular sample, while also being honest that the precise size of the effect carries real uncertainty.

5.4 Which tactical dimensions actually matter

The permutation importance analysis, measuring how much each tactical dimension actually contributes to the model's accuracy at matching players to their real managers, produced a clear and consistent ranking across every stage of the project. Pressing intensity, once properly possession adjusted, is by a wide margin the single most important dimension, and its importance grew substantially once the football analyst driven improvements were implemented, very nearly tripling from an already leading position. This confirms a specific, falsifiable prediction stated early in the project, that pressing would be the dimension most likely to statistically separate these managers, precisely because pressing is a defensive behaviour that does not depend heavily on a team's own technical possession quality, unlike most of the other dimensions used in this project. The separation this dimension actually produces, discussed in detail in the next section, cleanly confirms the predicted position of one manager, De Zerbi, but not the predicted ordering between the other two, which is itself a genuinely useful finding rather than a disappointing one.

Individual defending, which had previously shown no useful signal at all and in fact appeared to be actively adding noise rather than helping, became a genuinely useful dimension once the more detailed positional grouping was introduced, a clear demonstration that the earlier, broader positional categories had been obscuring real signal rather than simply failing to add any. Three further dimensions, buildup directness, progressive passing and progressive carrying, moved from a negligible contribution to a clearly positive one over the same improvements, broadening the set of dimensions doing real discriminating work well beyond pressing intensity alone. Box presence, by contrast, remains a mild net negative contributor even after the same improvements, meaning the more detailed positional grouping helped individual defending without doing the same for box presence, a useful reminder that not every dimension responds equally to the same fix, and that this should be stated plainly rather than implied to have improved uniformly. Chance creation also remains a mild net negative, suggesting that raw passing and creation based measures, even expressed as a share of a player's total activity rather than as a raw count, remain too closely tied to a player's individual technical quality and his team's overall possession dominance to reliably separate managerial philosophy from squad strength on their own.

5.5 The club identity check

The check comparing De Zerbi's own four clubs against each other, and against Guardiola's and Klopp's single clubs, is the project's most direct test of whether the model captures managerial philosophy or simply relearns which club a player belongs to. In the very first version of the model, before any corrections, this check produced the suspicious, extreme pattern already described, in which two of De Zerbi's clubs appeared almost the statistical opposite of every other club in the

dataset. After the coding and data errors were corrected, this extreme pattern disappeared, but a smaller, genuine gap remained, in which De Zerbi's own clubs resembled each other somewhat less than they resembled Guardiola's and Klopp's clubs, a result that pointed toward squad and league strength still dominating the underlying signal.

After the football analyst driven improvements described in the methodology section were implemented, this gap narrowed very substantially, by roughly a factor of six, to the point where De Zerbi's own clubs now resemble each other only very slightly less than they resemble the other two managers' clubs. This is genuine, measurable progress, and it is consistent with everything else the results show, that the specific, targeted improvements informed by football expertise did real work in separating managerial signal from squad and league strength, without fully eliminating that confounding effect.

The club identity check, before and after

- Before correction: De Zerbi's clubs looked like the statistical opposite of every other club, an extreme pattern later traced to a data error
- After fixing the bugs, before the analyst review: De Zerbi's own clubs resembled each other clearly less than they resembled the other managers' clubs
- After the football analyst driven improvements: the gap narrowed by roughly a factor of six, though a small difference remains

5.6 Clustering results

The clustering analysis across the full population of more than eleven thousand player seasons produced five interpretable groups, each named automatically from its two strongest tactical dimensions. These groups broadly correspond to recognisable football archetypes, including a defending and pressing oriented group, a build up and progressive passing oriented group, and a chance creation and progressive carrying oriented group, among others. The validity measures used to check whether this number of groups was appropriate consistently suggested that playing styles form a continuum rather than sharply separated categories, with the best available separation, measured by the silhouette score, sitting at a value that indicates real but modest structure. This is an honest and, in the context of football, entirely plausible finding, since very few professional analysts would claim that playing styles fall into a small number of perfectly distinct boxes rather than shading gradually into one another.

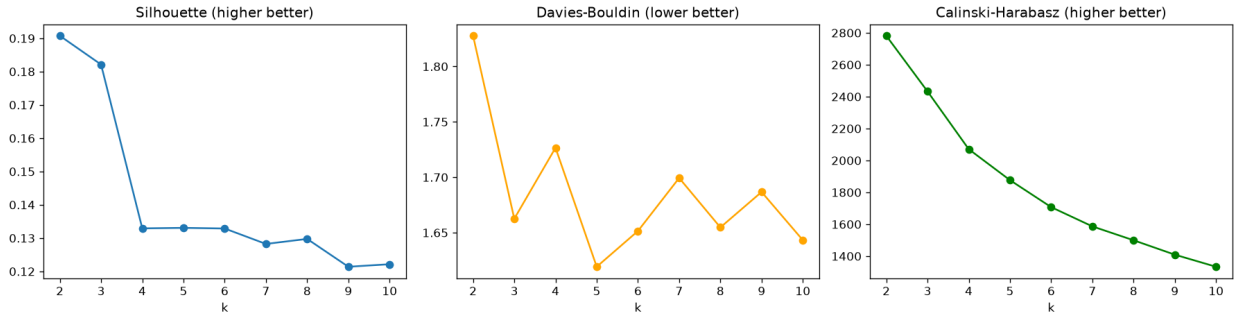


Figure 3: Silhouette, Davies Bouldin, and Calinski Harabasz validity scores across a range of candidate cluster counts. No number of clusters produces sharp separation, which matches the football intuition that playing styles form a continuum rather than distinct boxes.

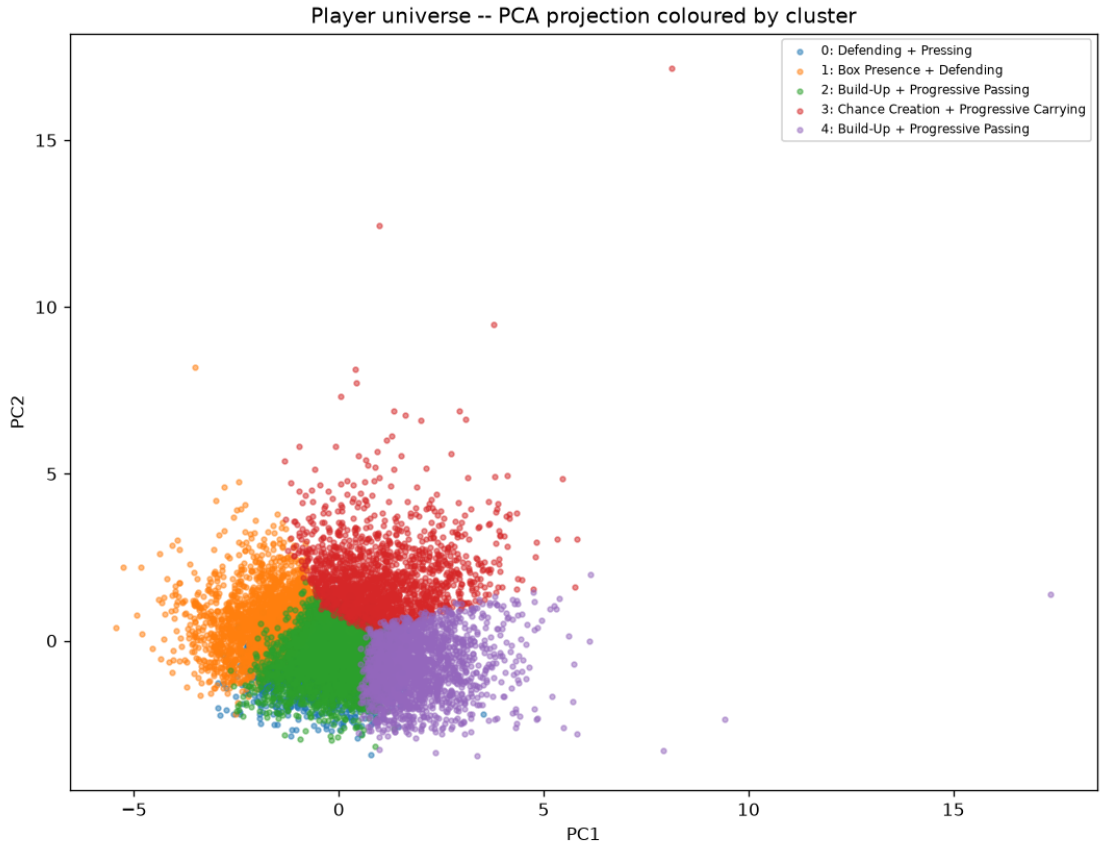


Figure 4: The full population of more than eleven thousand player seasons, projected onto two dimensions and coloured by its assigned playing style cluster. Each colour corresponds to one of the five named archetypes discovered by the clustering analysis.

5.7 A full worked example

To make the results described above concrete, it is worth presenting one complete example of the system's output exactly as a user would see it inside the application, using Erling Haaland's two thousand twenty two season at Borussia Dortmund as the case, since this example was used directly during development to sanity check that the whole explanation chain was behaving sensibly. Every image in this subsection is an actual screenshot of the running application, not a recreation, captured directly from the interface a user would open.

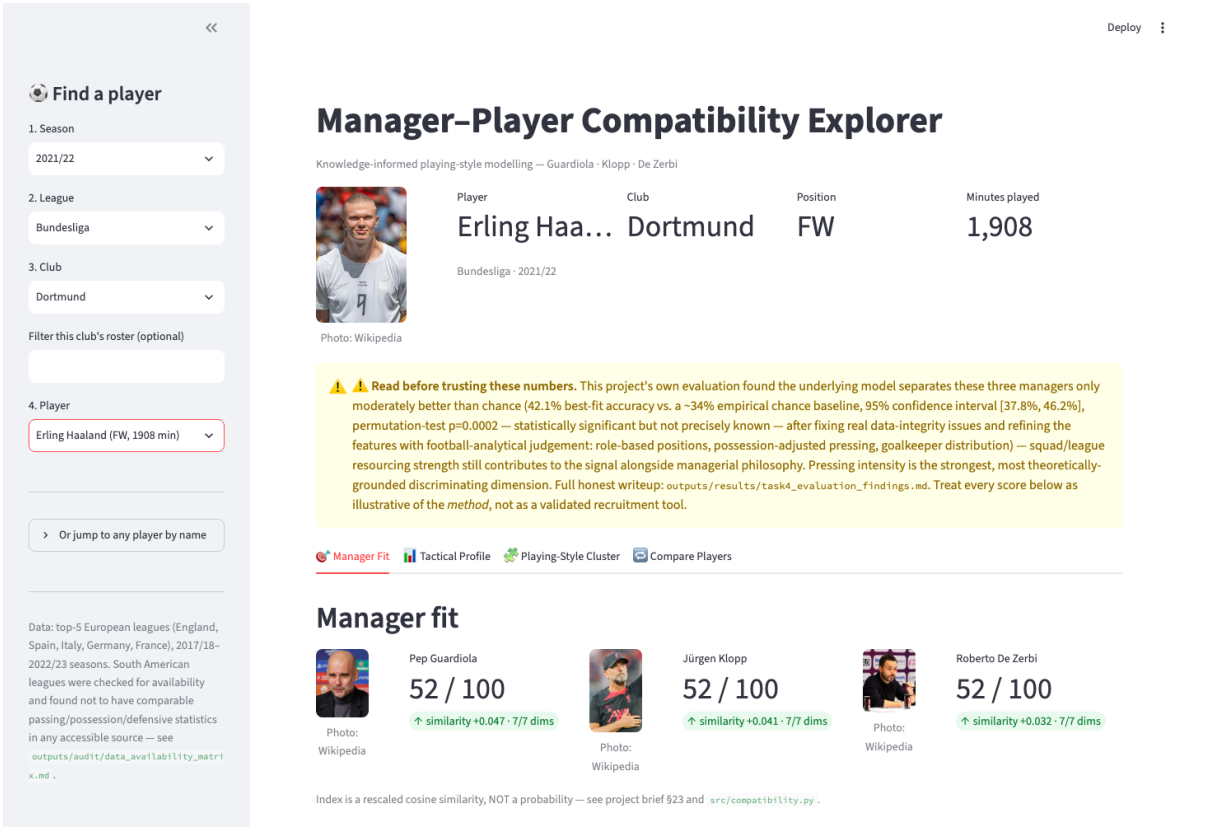


Figure 5: The application's own sidebar filters, cascading from season to league to club to player exactly as intended, with Erling Haaland's two thousand twenty two Dortmund season selected. The limitations warning banner is visible immediately, before any result, and every manager card carries its own photograph and its own similarity figure rather than a single blended score.

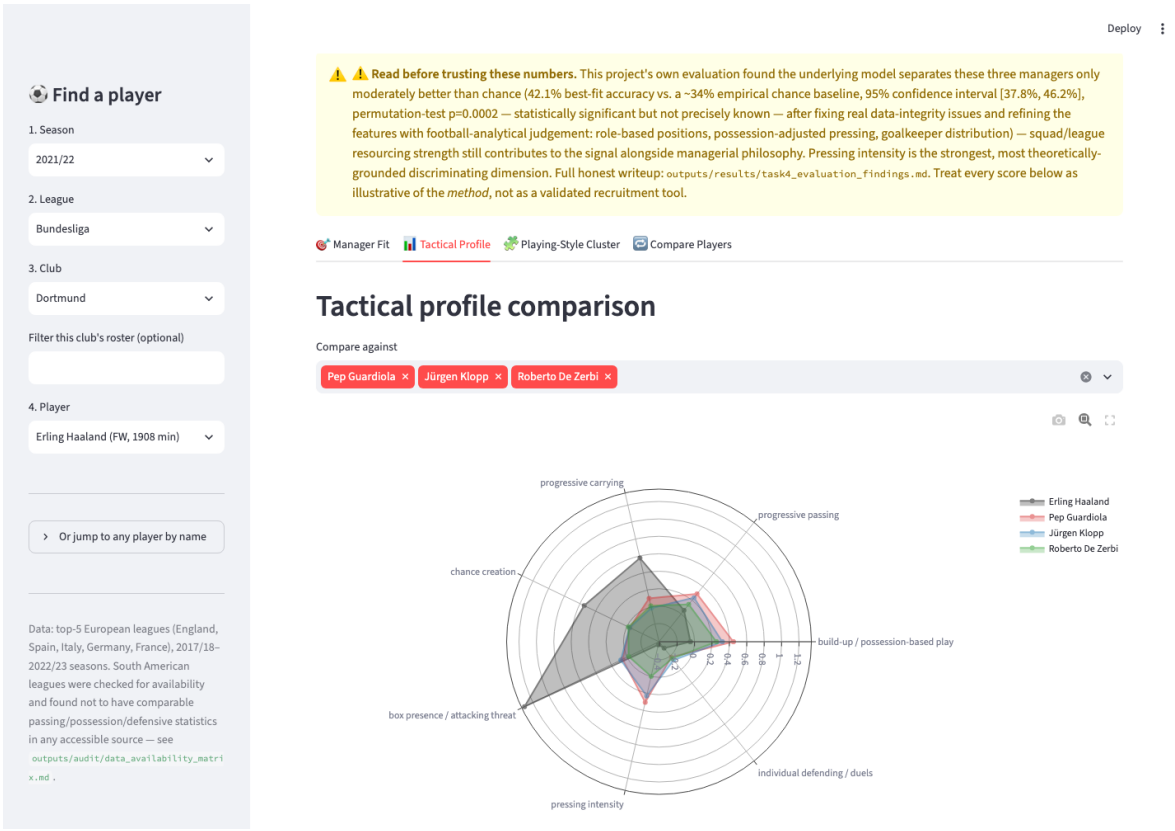


Figure 6: Erling Haaland's tactical profile compared directly against Guardiola, Klopp, and De Zerbi, screenshotted directly from the application's own interactive chart. The clear outlier on box presence and attacking threat is exactly the characteristic the system identifies as driving his strongest match.

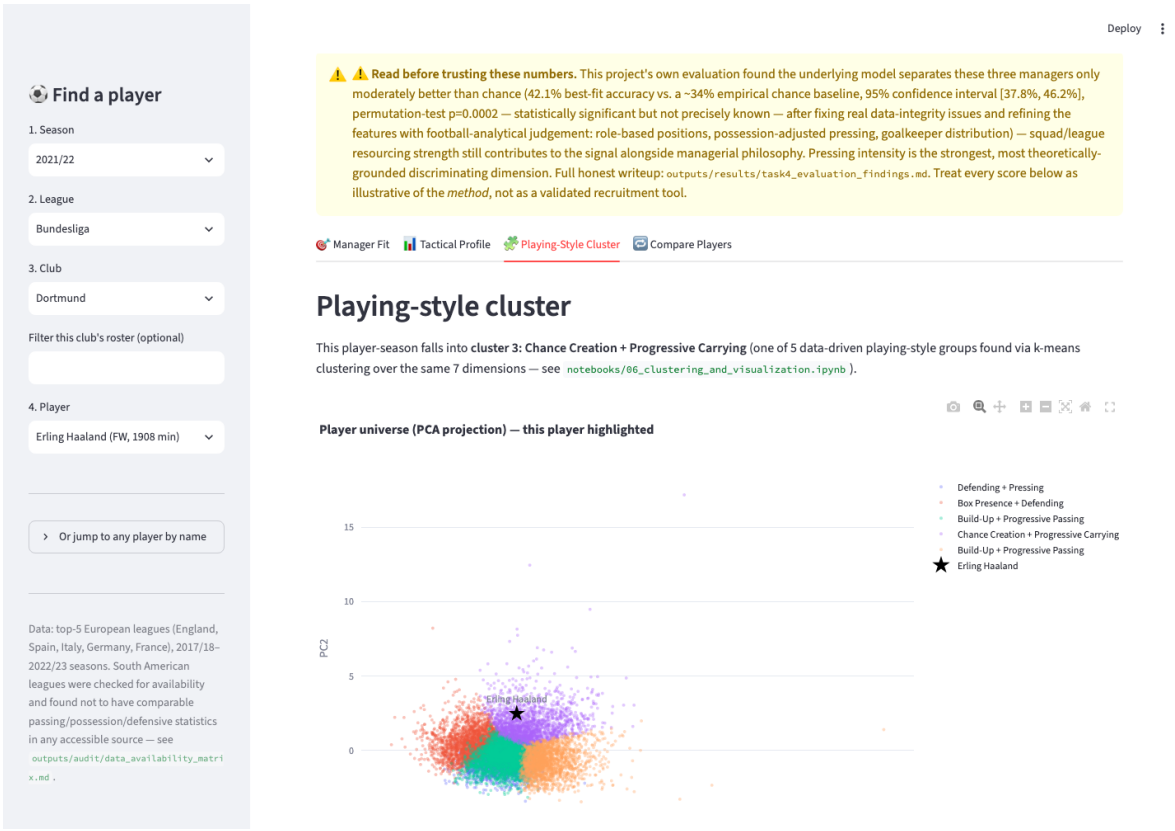


Figure 7: Where Erling Haaland sits within the full player universe once every player's tactical profile is projected onto two dimensions, screenshotted directly from the application. He falls right at the boundary of the chance creation and progressive carrying cluster, consistent with an elite penalty area goal threat whose broader profile is still recognisably that of a modern forward.

For this player and season, the system computed the highest similarity with Guardiola's profile among the three managers, ahead of both Klopp and De Zerbi, with all seven tactical dimensions available for the comparison, meaning the result carries no low confidence warning. When the system decomposed this result into its contributing parts, the strongest single contributor by a clear margin was box presence, on which this player scored far above the average for his position, reflecting an extremely high rate of touches and shots taken inside the penalty area relative to other forwards in the same population. The next two contributors, in order, were progressive carrying and individual defending, the latter appearing as a positive contributor despite the player himself scoring below the position average on it, specifically because his rate was closer to Guardiola's own team level tendency, whose possession dominant sides also need fewer defensive duels, than to the other two managers, illustrating that a contribution to similarity does not always mean a player scores unusually high on a dimension, only that his score sits closer to the manager in question than to the alternatives. On the other side of the ledger, the single largest factor working against an even stronger match was pressing intensity, on which the player scored below the position average in a way that pulled him further from Guardiola specifically, alongside a smaller pull from build up

involvement, consistent with a striker whose defensive workload off the ball and involvement in his team's own build up are both limited compared with his goal threat.

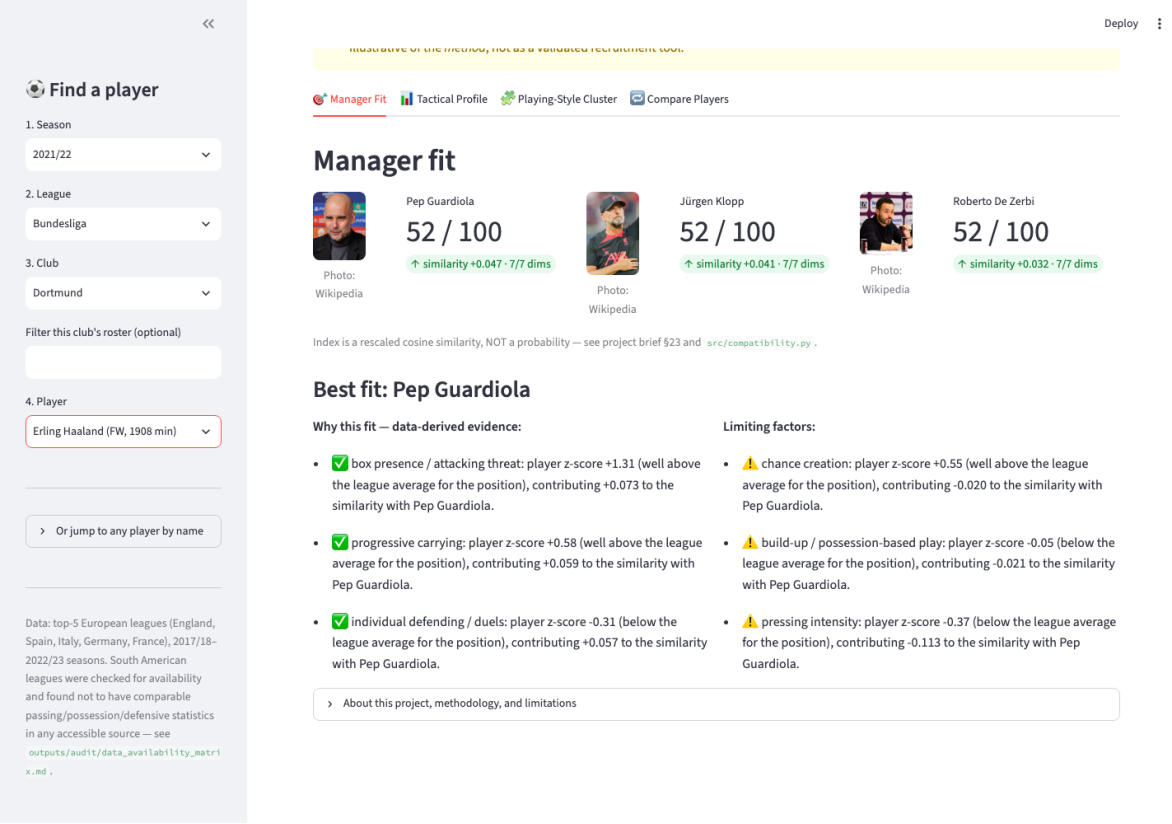


Figure 8: The explanation screen for this exact comparison, screenshotted directly from the application. Every figure quoted in the paragraph above, the three manager similarity scores and the six named contributors split into evidence for the fit and factors limiting it, is visible here exactly as a user would see it, with nothing added or paraphrased for the purpose of this report.

This example illustrates precisely what the explainability requirement set out in the objectives was meant to achieve. A user is never shown a bare number. They are shown a specific, checkable account of which measurable characteristics drove the result, expressed in terms that describe football behaviour rather than statistical machinery, with the option to inspect the exact underlying figures if they wish to challenge or verify the conclusion.

5.8 The application as a delivered result

Beyond the numerical results, the interactive application itself stands as a concrete, usable deliverable of the project. It was tested directly during development by searching for real, well known players, including a specific test using Erling Haaland, whose profile was correctly identified as showing an unusually strong box presence score for his position, leading to a sensible, explainable match with Guardiola's system, precisely the kind of coherent, checkable output the whole

explainability approach was designed to produce. The application successfully retrieves and displays real photographs for the great majority of well known players and for all three managers, correctly falls back to a placeholder for players without a suitable available photograph, and correctly displays its own limitations prominently to every user regardless of which player they choose to look at.

5.9 Team formation profiles: an unused signal, added descriptively

One of the recommended next steps listed later in this report, that a team's starting formation had been collected for every match since early in the project but never analysed, has now been explored, and the exploration itself is worth describing before the result.

The original plan was to fold formation into the compatibility score as an eighth tactical dimension, built the same way as the other seven. Checking the coverage of the live upstream match level source before doing so revealed a genuine complication. The source this project relies on for match level statistics publishes its data as a single, continuously updated file rather than a versioned snapshot for each season, and by the time this addition was attempted, that file had moved on to cover the current season and no longer contained the two thousand eighteen to two thousand twenty three window this project needs at all. The historical formation data was recovered successfully from an intermediate file saved much earlier in the project, which still held the original snapshot with formation information for every one of its rows. Nothing was actually lost, but only because this project happened to save every intermediate stage of its own pipeline rather than relying on being able to re download the same data again later. This is itself a small but genuine addition to the project's own record of data availability surprises, alongside the two described earlier in this report.

Having recovered the data, a second complication became clear. Formation information only exists for the twenty two club seasons inside the manager attributed sample used to build the compatibility score, and never for the much larger population of eleven thousand one hundred and thirty two player seasons that every other tactical dimension is standardised against. Building a numeric dimension out of only twenty two observations would be far more statistically fragile than any of the other seven dimensions in this project, so rather than quietly add a weak eighth dimension, the result is reported here descriptively instead, kept clearly separate from the compatibility score itself.

Manager	Matches	Modal formation	Stability	Mean back line
Jürgen Klopp	266	4 3 3	86.5 percent	4.00
Pep Guardiola	266	4 3 3	59.8 percent	3.87
Roberto De Zerbi	237	4 2 3 1	56.1 percent	3.81

Klopp is by a wide margin the most tactically settled of the three, playing his single most common formation in nearly nine of every ten matches across his entire covered tenure, consistent with a manager known for imposing one clear system and sticking with it rather than adapting it constantly. Guardiola experiments considerably more than his reputation for a single, clean possession iden-

tity might suggest, using ten distinct formations across his tenure and dropping to a back three often enough to pull his average defensive line size below four. De Zerbi is confirmed as the most structurally fluid of the three by every measure captured here, using the most distinct formations of the three and showing the lowest average defensive line size, which lines up closely with his description elsewhere in this report as an unconventional tactician whose signature approach operates through shifting shape while his own team has the ball rather than through a fixed structure. This is a genuinely new, independently corroborating piece of evidence for a claim this project was previously only able to make on qualitative grounds.

Pulling all of the above together, the honest summary of what this project's results show is the following. A knowledge informed system built from publicly available football statistics can separate the tactical philosophies of these three specific managers meaningfully better than random guessing, but not decisively so. The single strongest and most theoretically well grounded piece of evidence the system finds is that Klopp's teams press with an intensity, once properly adjusted for how much possession his teams actually have, that clearly exceeds both Guardiola's and De Zerbi's teams, exactly matching independent football analysis of his career. The system also finds real, if less dramatic, evidence that De Zerbi's teams generate their attacking value through in possession mechanisms rather than through pressing, again matching independent football analysis of his specific tactical signature. What the system does not yet reliably separate, despite substantial and successful effort to improve it, is the more general question of how much possession a team retains and how it progresses that possession through passing, because these particular measures remain too closely entangled with simple squad quality and league strength to cleanly isolate managerial intent using the kind of aggregated match statistics available for this project.

6. Conclusion and Next Steps

6.1 Answering the research question

The research question that opened this report asked whether football domain knowledge and player performance data can be combined with machine learning to model the playing philosophies of elite managers, and whether the result can estimate player compatibility in an explainable way. Based on everything described in this report, the honest answer has two distinct parts.

On the question of explainability, the answer is a clear yes. Every compatibility score produced by this project's final system can be broken down into the specific tactical characteristics that drove it, expressed in language a football person would recognise, with the underlying statistical evidence always available for inspection. This was achieved consistently throughout the project and was never compromised for the sake of a more impressive looking but less transparent result.

On the question of modelling accuracy, the answer is a qualified yes. The final system separates these three specific managers meaningfully better than chance, led convincingly by pressing intensity, a dimension whose importance and direction closely match independent football analysis conducted earlier in the project. This is a real, demonstrated result, not a null one, and a permu-

tation test described in section 5.3 confirms it directly, with a p value of zero point zero zero zero two against an empirical chance baseline of thirty four percent. At the same time, the overall accuracy achieved, just over forty two percent against a one in three baseline, carries a ninety five percent confidence interval spanning from thirty seven point eight to forty six point two percent, wide enough that the point estimate should not be read as more precise than it is. It is a genuine and useful improvement over the earlier, chance level result, but it falls well short of the kind of accuracy that would justify treating this system as a finished, reliable recruitment tool rather than as a promising research demonstration.

6.2 What made this project's evidence trustworthy

The single most valuable methodological lesson from this project is that an unusually clean, dramatic result should always be treated with suspicion rather than celebration until it has survived a genuine attempt to break it. The first version of the compatibility model produced a striking, almost perfect looking pattern separating De Zerbi's clubs from everyone else, and it would have been easy, and would have made for a more impressive sounding early result, to report that pattern as a finding. Investigating it instead revealed two real, serious errors, a bug in how missing data was handled and a previously undiscovered gap in the underlying data source, both of which were corrected before any conclusion was drawn from the results. The far more modest result that emerged after those corrections, and the subsequent, carefully targeted improvements that raised accuracy in a way that could be checked and explained rather than simply asserted, is a more defensible piece of work precisely because of, rather than despite, that early disappointment.

6.3 Limitations, stated plainly

This project has several limitations, and they are stated here plainly rather than left implicit.

> ⚠ Read before trusting these results. This system separates the three managers only moderately better than chance, 42.1 percent accuracy (95 percent confidence interval 37.8 to 46.2 percent, permutation test p equals 0.0002) against an empirical chance baseline of 34 percent. Squad and league resourcing strength still contributes to the signal alongside managerial philosophy. This is a research demonstration, not a validated recruitment tool.

The available public data covers aggregated match statistics rather than positional tracking information, which means the system can see the outcomes of a manager's tactical structure, such as how many progressive passes a player completes, but cannot directly see the structure itself, such as the shape of the team or the spacing between players. This is the most fundamental limitation of the project, and it is almost certainly the main reason the model's accuracy, while genuinely improved, still falls short of anything decisive.

Guardiola and Klopp's available data comes from only one club each within the period covered by the data source used, Manchester City and Liverpool respectively, while De Zerbi's data spans four different clubs. This makes the club identity check described earlier possible for De Zerbi but

not for the other two managers, and it means the project cannot fully rule out that some of what looks like Guardiola's or Klopp's signature is actually a reflection of the specific, well resourced clubs they happened to manage during the period the data covers.

The evaluation approach used throughout the project, checking whether a player's best statistical match is the manager he actually played under, is a demanding and useful test, but it is not a perfect one, because players are not freely assigned to clubs purely on the basis of tactical fit. Contracts, transfer fees, and existing squad needs all play a role in determining who actually plays for whom, which means even a hypothetically perfect model of tactical philosophy would not be expected to achieve one hundred percent accuracy on this specific measure.

South American leagues were checked directly and found not to be covered by any accessible data source with sufficient statistical detail, and are therefore absent from the final player database and application, a limitation stated clearly to the user rather than left unexplained.

Photographs displayed in the application are retrieved automatically by matching a player's name against Wikipedia, and while this lookup is deliberately conservative and falls back to a placeholder when it cannot be confident of a match, it is not guaranteed to be correct in every single case, particularly for players who share a name with someone else.

6.4 Recommended next steps

Several concrete directions for future work follow naturally from the limitations just described, and from ideas identified but not pursued during this project owing to time constraints.

The most promising single direction is to explicitly account for a player's own team strength when computing his tactical dimension scores, comparing a player not only to the wider league population in his position but also to his own teammates, which would directly target the remaining confusion between managerial philosophy and squad quality that this report has described as the project's central limitation.

A second promising direction is to move beyond aggregated match statistics toward positional tracking data, which would make it possible to build features that directly describe team structure, such as the shape formed by outfield players during build up or the sequence and tempo of passing moves, rather than only the outcomes of that structure. This would require a different, likely more restricted data source, but it would directly address the fundamental limitation identified in this report.

A third direction, requiring no new data at all, is to build separate manager profiles for each tactical role rather than one single blended profile per manager, since part of what defines a manager's identity is precisely which roles he chooses to emphasise, something that a single averaged profile across every position on the pitch necessarily obscures.

A fourth direction is to explicitly adjust every tactical dimension for the specific league a player competes in, rather than pooling all five leagues into a single comparison population, since different leagues are known to differ in overall tempo and physicality in ways that could otherwise be

mistaken for managerial signal.

A fifth and more modest direction, achievable quickly if this project were continued, is to close the one clearly identified remaining gap in the project's own qualitative research, a thinner than desired investigation into De Zerbi's specific out of possession and defensive tactics, which would strengthen the football grounded interpretation of his profile.

A sixth direction, using the starting formation recorded for every team in every match, has now been explored, described fully in section 5.9. The formation data itself held up well, and the resulting manager profiles are genuinely informative and independently corroborate this report's qualitative description of De Zerbi as the most tactically fluid of the three managers. It was not folded into the compatibility score as an eighth dimension, because formation is only available for the twenty two club seasons in the manager attributed sample, far too small a population to standardise safely alongside the other seven dimensions, each of which draws on the full eleven thousand plus player season population. Closing that remaining gap, so that formation could be added as a proper eighth dimension rather than a descriptive supplement, would require either locating a data source with formation coverage across the full player population, or an explicit statistical treatment suited to a small sample, and remains a genuine direction for future work rather than something this project could complete within its own timeline.

A seventh direction concerns the pace and rhythm of possession itself, rather than only its share and its outcomes. The project's own research into these three managers repeatedly referenced measures used by professional analysts such as the number of passes completed within a single unbroken spell of possession, and the average speed at which a team moves the ball toward the opponent's goal once it wins possession back. None of the statistics available to this project captured tempo or sequence length directly, and building features of this kind, even in an approximate form, would target exactly the sequential, structural aspect of style that this report has repeatedly identified as the hardest thing for aggregated match statistics to see.

Finally, and this is stated as a firm recommendation rather than a suggestion, no version of this system as it currently stands should be presented to a club or any other decision maker as a validated recruitment tool. It should continue to be described, exactly as it is described inside the application itself, as a research demonstration with clearly stated accuracy and clearly stated limitations, until further work of the kind described above has been carried out and independently evaluated.

6.5 What this project actually contributes

It is worth stating plainly, in one place, what this project adds to what already existed before it, since a report of this length can easily leave a reader uncertain about which parts are genuinely new and which parts build on established ideas.

The individual pieces of statistical technique used in this project, principal component style reasoning about which statistics belong together, z scoring against a comparison population, cosine similarity between profiles, k means clustering, and permutation importance, are all well estab-

lished and none of them is presented here as a novel technique in its own right. What this project contributes instead is the specific combination of these established techniques into a single, coherent, and thoroughly documented pipeline aimed at a question the literature reviewed for this project does not appear to answer directly, namely whether an individual manager's philosophy, tracked across different clubs where the data allows it, can be compared against an individual player in a way that is both measured against real data and explained in football language rather than left as an opaque score. The explicit football knowledge layer, separating raw statistics from football concepts and football concepts from tactical dimensions, with every step documented and every limitation stated, is the specific structural contribution this project makes, together with the fully worked, end to end demonstration of that structure inside a real, usable application.

Equally, this project contributes a documented account of a specific failure mode that any similar future project should watch for, the way a genuine data gap combined with a subtle coding error can produce a result that looks like a strong, exciting finding about football when it is really an artefact of missing information. Very few project reports describe this kind of problem openly, since there is an understandable temptation to present only the final, corrected pipeline as though it had always worked that way. This report has instead described the mistake, how it was found, and exactly what was done to fix it, because that account is itself a useful contribution to anyone building a similar system in the future.

6.6 Closing reflection

This project set out to answer a genuinely difficult question, whether a manager's footballing philosophy can be captured from public data well enough to say, in explainable football language, how well a given player would fit that system. The honest answer produced by rigorous, repeatedly self checked work is that this is possible to a real but limited degree, that the single clearest success is pressing intensity, correctly and independently identified as the dimension that separates the three managers most clearly, with De Zerbi's distinctly lower score confirming exactly what the project's own qualitative research predicted for him, even though the specific comparison between Guardiola and Klopp on that same dimension did not match the popular narrative and turned out to be informative for a different reason, a residual squad strength effect rather than a managerial one. The project's greatest strength is not any single number it produced but the discipline with which it questioned its own early, flattering results, found genuine errors, corrected them, and reported what was left afterward without exaggeration, including this specific correction. That discipline, more than any particular accuracy figure, is what makes this project's conclusions worth trusting.